UTMxHackathon'26

Official Case Study

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ISSUED BY

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Welcome to UTMxHackathon'26!

Case Study 1:
Precision Urban Agriculture for Vertical Farming
(IoT System/ Web Dashboard/ Mobile Application)

1. **Introduction**

   Rapid urbanization, shrinking arable land, and climate change are putting immense pressure on traditional food supply chains. In densely populated areas, food security and the carbon footprint of transporting produce are growing concerns. Vertical farming offers a space-efficient, sustainable alternative by growing crops in stacked layers within controlled indoor environments, making it possible to produce food directly within city limits

2. **Problem Statement**

   Managing a vertical farm manually is highly resource-intensive and prone to human error. Inconsistent lighting, improper nutrient delivery, and fluctuating climate control can quickly lead to suboptimal crop yields, crop disease, wasted water, and excessively high energy consumption. Currently, many urban farmers and hobbyists lack access to affordable, data-driven automation tools needed to precisely monitor and optimize plant growth around the clock.

3. **Challenge**

   Develop an intelligent, automated system that monitors and manages the environmental variables of an indoor vertical farming setup in real-time. The platform should collect sensor data, automate routine agricultural tasks, and provide actionable, data-driven insights to help users maximize crop yield while strictly minimizing water and electricity usage. It should be scalable enough for a small community garden or an indoor campus greenhouse.

4. **What the solution should solve**

   Participants can build a working hardware prototype (optional) paired with a software dashboard. Key features could include:

   - Real-time monitoring using sensors for temperature, ambient humidity, soil moisture, and water pH levels.
   - Automated control systems to adjust LED lighting spectrums, activate cooling fans, or trigger hydroponic water pumps based on specific plant profiles.
   - A centralized web or mobile dashboard that visualizes growth metrics, resource consumption, and historical data trends.
   - Predictive alerts that notify users of system anomalies, such as a water reservoir running dry or temperatures exceeding safe thresholds.
   - An AI-driven recommendation engine that suggests optimal harvest times or adjustments to nutrient mixtures based on sensor inputs.

Area	Tools
Hardware / IoT	Arduino, ESP32, Raspberry Pi, Soil Moisture/pH Sensors
Backend	Node.js, Python (Flask/FastAPI), Firebase, Supabase
Frontend	React, Vue.js, Flutter, React Native
Data/AI	TensorFlow Lite, Scikit-learn, OpenAI API for crop insights
Communication	MQTT, WebSockets, REST APIs

The solution should be:

- A functional hardware (optional) and software prototype demonstrating real-world data collection.
- Interactive and user-friendly, allowing individuals without deep agricultural expertise to manage a system.
- Scalable to adapt to different types of crops or larger vertical farming racks

Note: The features and tools listed above are provided as guidance only. Participants are strongly encouraged to explore alternative approaches that meaningfully address the problem statement.